LAB 1: GRADIENT DESCENT AND STOCHASTIC GRADIENT DESCENT

Assessment type: laboratory

Assessment weighting: 6% of total module mark

Module learning outcomes assessed by this task:

- CLO 3. Be able to develop computer programs that apply iterative algorithms to solve optimization problems in various applications.

Using the training and testing datasets generated as in https://www.cvxpy.org/examples/machine_learning/ridge_regression.html with the objective

$$\min \frac{1}{m} ||X\beta - y||_2^2 + \lambda ||\beta||_2^2$$

$X \in R^{m \times n}$, $\lambda > 0$.

Step 1. Use cvx solver to find the solution of linear regression with mean square error and norm-2 regularization.

Show the effect of coefficient λ on the train and test errors on the variables.

Step 2. Write the code of the gradient descent algorithm to solve the optimization problem (with a particular coefficient λ).

Compare the solution to the one solved by cvxpy.

Step 3. Show the stochastic gradient descent algorithm solving the optimization problem.

Write the code and compare it to the solution solved by cvxpy.

Step 4. Write the code applying the batch gradient descent algorithm to solve the problem with different batch sizes.

Step 5. Compare and contrast the results from different algorithms.

Rubrics for lab assessment:

Criteria	8-10	5-7	4-0
Codes (60 points)	The codes run correctly for all steps, the iterative algorithms converge to the optimal solution; Plots demonstrate	1-2 steps haven't been finished; The code is correct, however, the algorithms do not converge and some plots are in correct.	Just few steps have been done.

	the results clearly		
Presentation (40 points)	Demonstrates full knowledge of the topic, the report is well organized	The content shows some understandin g and comprehension, few steps haven't been finished	The presentation has some information about the topic, but is mostly based on clichés and basic knowledge.